

LAB ASSIGNMENT - 3

Shell Scripts

1. To find Largest of Three Numbers.

Solution:

```
echo "Enter three numbers:"
read a b c
if [ $a -ge $b ] && [ $a -ge $c ]; then
    echo "$a is the largest"
elif [ $b -ge $a ] && [ $b -ge $c ]; then
    echo "$b is the largest"
else
    echo "$c is the largest"
fi
```

2. To find a year is leap year or not.

Solution:

```
echo "Enter year:"
read year
if (( year % 400 == 0 || (year % 4 == 0 && year % 100 != 0) )); then
    echo "$year is a leap year"
else
    echo "$year is not a leap year"
fi
```

3. To input angles of a triangle and find out whether it is valid triangle or not

Solution:

```
echo "Enter three angles:"
read a b c
sum=$((a + b + c))
if [ $sum -eq 180 ] && [ $a -gt 0 ] && [ $b -gt 0 ] && [ $c -gt 0 ]; then
    echo "Valid triangle"
else
    echo "Invalid triangle"
fi
```

4. To check whether a character is alphabet, digit or special character.

Solution:

```
echo "Enter a character:"
read ch
if [[ $ch =~ [A-Za-z] ]]; then
    echo "Alphabet"
elif [[ $ch =~ [0-9] ]]; then
    echo "Digit"
else
    echo "Special character"
fi
```

5. To calculate profit or loss.

Solution:

```
echo "Enter Cost Price and Selling Price:"
read cp sp
if [ $sp -gt $cp ]; then
    echo "Profit of $((sp - cp))"
elif [ $cp -gt $sp ]; then
    echo "Loss of $((cp - sp))"
else
    echo "No profit no loss"
fi
```

6. To print all even and odd number from 1 to 10

Solution:

```
echo "Even numbers:"
for i in {1..10}; do
    if (( i % 2 == 0 )); then echo $i; fi
done

echo "Odd numbers:"
for i in {1..10}; do
    if (( i % 2 != 0 )); then echo $i; fi
done
```

7. To print table of a given number

Solution:

```
echo "Enter a number:"
read n
for i in {1..10}; do
    echo "$n * $i = $((n * i))"
done
```

8. To find factorial of a given integer

Solution:

```
echo "Enter a number:"
read n
fact=1
for ((i=1; i<=n; i++)); do
    fact=$((fact * i))
done
echo "Factorial of $n is $fact"
```

9. To print sum of all even numbers from 1 to 10.

Solution:

```
sum=0
for i in {1..10}; do
    if (( i % 2 == 0 )); then
        sum=$((sum + i))
    fi
done
echo "Sum of even numbers: $sum"
```

10. To print sum of digit of any number.

Solution:

```
echo "Enter a number:"
read n
sum=0
while [ $n -ne 0 ]; do
    digit=$((n % 10))
    sum=$((sum + digit))
    n=$((n / 10))
done
echo "Sum of digits: $sum"
```

11. To make a basic calculator which performs addition, subtraction, Multiplication, Division

Solution:

```
echo "Enter two numbers:"
read a b
echo "Choose operation (+ - * /):"
read op
case $op in
    +) echo "$a + $b = $((a + b))" ;;
    -) echo "$a - $b = $((a - b))" ;;
    \*) echo "$a * $b = $((a * b))" ;;
    /) echo "$a / $b = $((a / b))" ;;
    *) echo "Invalid operation" ;;
esac
```

12. To print days of a week.

Solution:

```
for day in Monday Tuesday Wednesday Thursday Friday Saturday Sunday; do
    echo $day
done
```

13. To print starting 4 months having 31 days.

Solution:

```
months=(January March May July August October December)
echo "First 4 months with 31 days:"
for ((i=0; i<4; i++)); do
    echo ${months[$i]}
done
```

14. Using functions,

a. To find given number is Armstrong number or not

Solution:

```
is_armstrong() {
    num=$1
    sum=0
    temp=$num
    while [ $temp -gt 0 ]; do
        digit=$((temp % 10))
        sum=$((sum + digit * digit * digit))
        temp=$((temp / 10))
    done
    if [ $sum -eq $num ]; then
        echo "$num is an Armstrong number"
    else
        echo "$num is not an Armstrong number"
    fi
}
echo "Enter number:"
read n
is_armstrong $n
```

b. To find whether a number is palindrome or not

Solution:

```
is_palindrome() {  
    num=$1  
    rev=0  
    temp=$num  
    while [ $temp -gt 0 ]; do  
        digit=$((temp % 10))  
        rev=$((rev * 10 + digit))  
        temp=$((temp / 10))  
    done  
    if [ $rev -eq $num ]; then  
        echo "$num is a palindrome"  
    else  
        echo "$num is not a palindrome"  
    fi  
}  
echo "Enter number:"  
read n  
is_palindrome $n
```

c. To print Fibonacci series upto n terms

Solution:

```
fibonacci() {  
    a=0  
    b=1  
    echo "Fibonacci series up to $1 terms:"  
    for ((i=0; i<$1; i++)); do  
        echo -n "$a "  
        fn=$((a + b))  
        a=$b  
        b=$fn  
    done  
    echo  
}  
echo "Enter number of terms:"  
read n  
fibonacci $n
```

d. To find given number is prime or composite

Solution:

```
is_prime() {  
    num=$1  
    if [ $num -le 1 ]; then  
        echo "$num is neither prime nor composite"  
        return  
    fi  
    for ((i=2; i*i<=num; i++)); do  
        if (( num % i == 0 )); then  
            echo "$num is composite"  
            return  
        fi  
    done  
    echo "$num is prime"  
}  
echo "Enter number:"  
read n  
is_prime $n
```

e. To convert a given decimal number to binary equivalent

Solution:

```
to_binary() {  
    num=$1  
    bin=""  
    while [ $num -gt 0 ]; do  
        bin=$((num % 2))$bin  
        num=$((num / 2))  
    done  
    echo "Binary: $bin"  
}  
echo "Enter decimal number:"  
read n  
to_binary $n
```